

Data Scientist Assessment

Fintech Transaction Analysis & Fraud Detection

May 2024

Executive Summary

Key Metrics Overview

3,050	Total Transactions	200	Unique Customers
3	Month Period	78.95%	Success Rate
2.16%	Chargeback Rate	96.89%	Model Accuracy

✓ 0 missing values | ✓ 0 duplicates | ✓ 4 data transformations

Problem Statement

Critical Findings:

Transaction Failure Rate: 21.05% (642 transactions failing)

Chargeback Risk: 2.16% (2-4x industry average of 0.5-1%)

High-Risk Transactions: 153 high-amount transactions >95th percentile

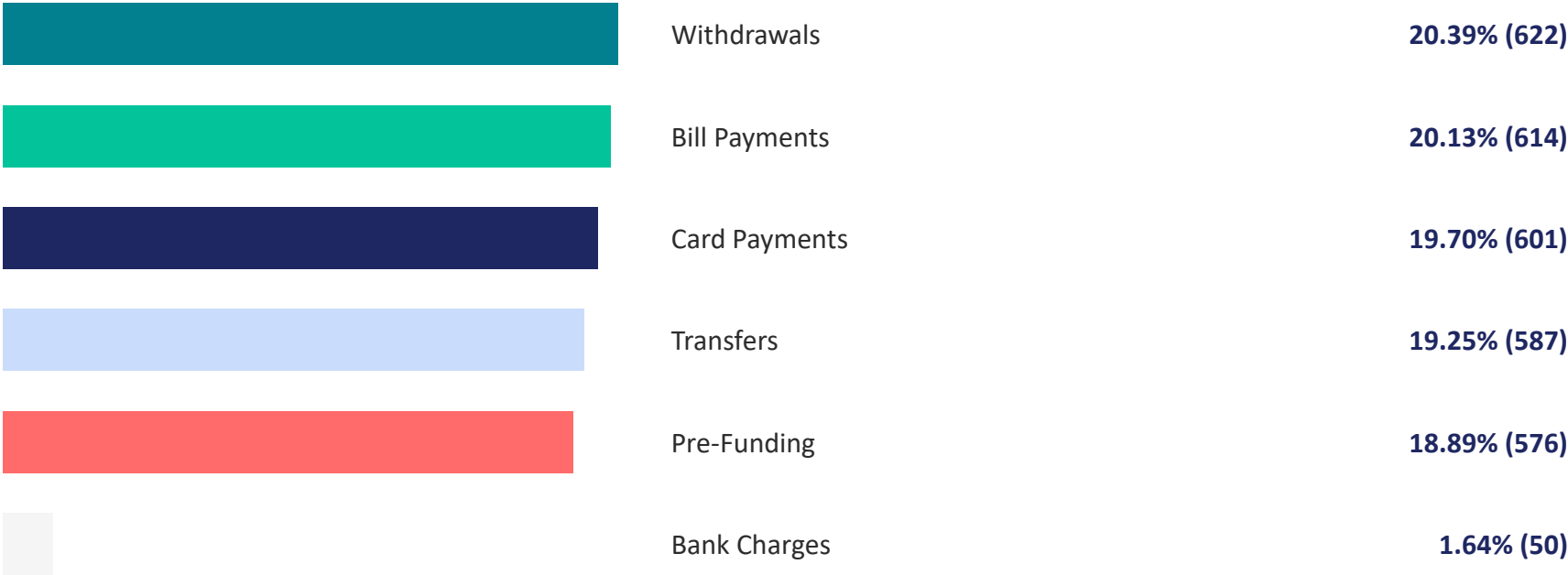
Pending Backlog: 320 transactions stuck in pending status

Data Quality Assessment

✓	Missing Values	0 found
✓	Duplicate Rows	0 found
✓	Data Integrity	Confirmed
✓	Date Range	Jan 1 - Mar 1, 2024
✓	Classifications	15 variations standardized to 6
✓	Transformations	4 major data cleaning actions

Transaction Analysis

Distribution by Classification Type



Transaction Status Breakdown

78.95%

Completed (2,408)

21.05%

Failed (642)

Pending: 320 (10.49%)

Cancelled: 161 (5.28%)

Refunded: 95 (3.11%)

Chargeback: 66 (2.16%)

Anomalies & Risk Detection

Risk Indicators

- Chargeback: 2.16%
- High-Risk Customers: 59
- Fraud Customers: CUST0038
- High-Amount Txns: 153
- Unusual 1AM Peak: 143/hr

Failure Rate by Category

- ⚠ Card Payments: 22%
- ⚠ Withdrawals: 22%
- ⚠ Transfers: 21%
- ⚠ Pre-Funding: 20%
- ⚠ Bill Payments: 20%

Feature Engineering

19 Features Created Across 5 Categories



Temporal (6)

Day of week, hour, is_weekend, is_high_hours, etc.



Customer Behavior (3)

Avg amount, txn count, chargeback rate



Amount-Based (4)

Log, scaled, is_high_amount, is_low_amount



Description-Based (3)

Is_transfer, is_bill_payment, is_withdrawal



Encoded Features (3)

Day of week, status, and other categorical encodings

Predictive Models: Performance Comparison

Random Forest

Accuracy	89.84%
Precision	92.37%
Recall	89.84%
F1-Score	90.11%

Gradient Boosting ★

Accuracy	96.89%
Precision	96.90%
Recall	96.89%
F1-Score	96.87%

+7.05% improvement

Feature Importance (Gradient Boosting)

Description-based features dominate (78% combined importance)



High-Priority Recommendations (1-2 Weeks)

Real-Time Fraud Detection

Action: Deploy Gradient Boosting model

#1 Priority

Real-Time Fraud Detection

Customer Risk Scoring

Action: Create composite risk metric

#2 Priority

Customer Risk Scoring

Enhanced Verification

Action: 2FA for transactions >15,000

Benefit: 25% fraud reduction (ROI: High)

#3 Priority

Enhanced Verification

7 Key Business Insights

1. Transaction pipeline has 21% failure rate - significant optimization opportunity
2. Chargeback rate (2.16%) is 2-4x higher than industry average
3. Card Payments and Withdrawals show highest failure rates (22%)
4. Description-based features account for 78% of model importance
5. Gradient Boosting model achieves 96.89% accuracy - production-ready
6. 320 pending transactions indicate process bottlenecks requiring urgent attention
7. Customer segmentation opportunity with highly varied transaction patterns

Questions?

Complete Analysis Available in Accompanying Documents

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